

SwachSetu – Digitized waste management

- Problem Statement ID -25060
- Problem Statement Title- Real Life Solution for waste management
- Theme- Clean and Green Technology
- PS Category- Software
- Team ID-
- Team Name- pillowmasters

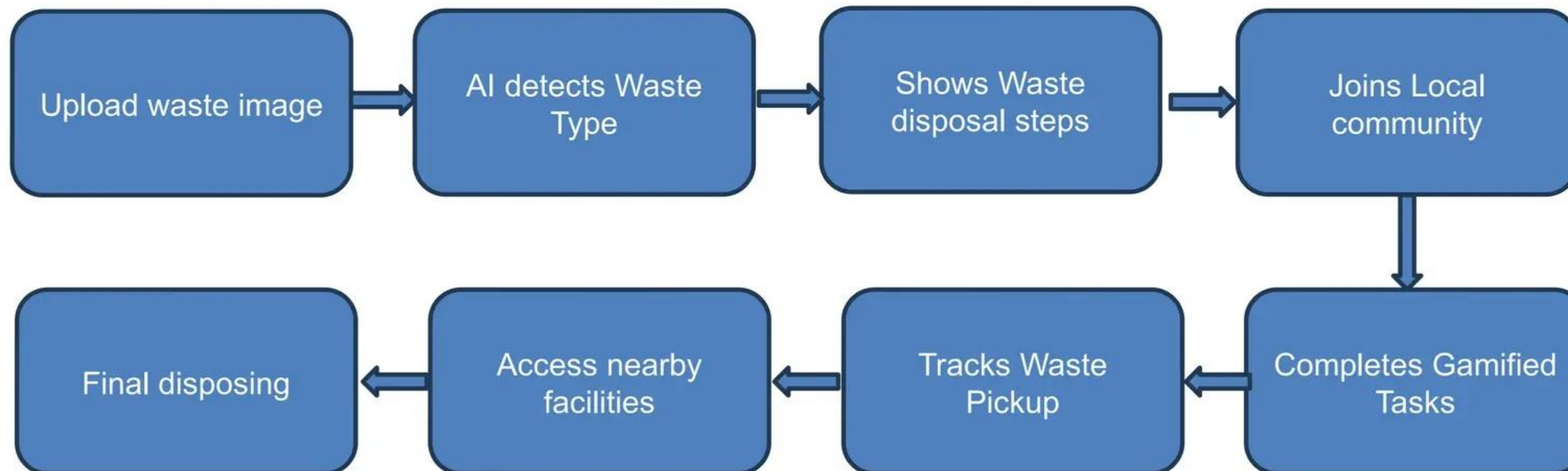


Complete Digitized Experience

❖ Proposed Solution

- AI Models detect dry/wet waste from images
- Web + Mobile App for training and waste tracking, and access to waste tool
- Geo Based Communities (Green Heroes) for local-level monitoring
- Gamified Tasks for improved citizen awareness
- End-to-End Waste Management guidance via AI-ML model
- Unique Combination of AI, Gamification and decentralized participation.

- Technologies to be used - NextJS, LLM, Supabase, PostgreSQL, RAG Machine Learning (NLP)



- Technically feasible using existing AI and app development frameworks.
- Operationally fits with ULB/GP workflows and national waste policies.
- Challenge: Low digital literacy may limit adoption in rural areas.
- Solution: Use simple UI and support multiple regional languages.
- Challenge: AI may misclassify mixed or unclear waste images.
- Solution: Improve AI with real data and allow manual correction.

IMPACT AND BENEFITS

- Empowers citizens with knowledge and tools for responsible waste disposal.
- Increases engagement through gamification and community participation.
- Reduces unsegregated and unaccounted waste at the source.
- Lowers landfill use, improving air, water, and soil quality.
- Creates economic opportunities via recycling, composting, and local waste markets.
- Strengthens decentralized waste monitoring and accountability in ULBs/GPs.

- <https://www.sciencedirect.com/science/article/pii/S0956053X09001275/>
- Baird Jim, The concept of Waste Management
- Alliance to end Plastic Waste, Amita Bhaduri , India Water Portal
- www.pgportal.gov.in (CPGRAMS)